

Ishwor Thapa, Ph.D., MS

[Portfolio](#) | [LinkedIn](#)

Biomedical Informatics | Computer Science | Bioinformatics Core Personnel

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Professional Summary

Results-driven Software Developer and Researcher with over 16 years of experience in the Bioinformatics Core at the University of Nebraska at Omaha. Expert in developing scalable Python applications, high-performance computing (HPC) pipelines, and agent-based models to solve complex biomedical challenges. Strong record of collaboration with premier institutions, including the Mayo Clinic and the University of Massachusetts, in clinical biomarker identification for prostate cancer and immune repertoire sequencing analysis. Authored 45+ research publications through extensive cross-institutional collaborations. Earned Coursera certifications in machine learning, genomic and precision medicine, virology and computational biology fields, alongside coursework in Molecular Biology of the Cell, reflecting interdisciplinary training in CS and life sciences.

Professional Experience

Software Application Developer | Bioinformatics Core Facility, UNO | 10/2009 – Present

- Developed and optimized bioinformatics pipelines for high-throughput omics data analysis, supporting researchers at multiple institutions.
- Configured high-performance computing (HPC) environments and managed Linux servers for complex biological simulations and data processing.
- Collaborated on grant-funded projects (NIH, Mayo Clinic) as key technical personnel, contributing to over 45 peer-reviewed publications.
- Designed hands-on modules for Bioinformatics analysis including RNA-Seq analysis, data visualization, and microbiome analysis.

Research Collaborator | Mayo Clinic, Dept. of Urology | 04/2025 – Present

- Leading the development of computational methods to integrate multi-omics data from extracellular vesicles for non-invasive prostate cancer biomarker discovery.
- Worked on the calibration, standardization and automation of nano-flow cytometry platform for robust and reproducible measurement of extracellular vesicles.
- Implementing machine learning models to predict clinical outcomes in oligometastatic cancer patients.

Software Developer | Sambad Project (Nepal) | 01/2006 – 08/2007

- Engineered computer interfaces specifically designed for non-literate users, focusing on human-computer interaction (HCI) accessibility.

Key Projects

Computational biomarker discovery

- Developed various computational methods including multi-omics integration for identification of biomarkers.
- Worked on developing a novel method for discovering potential multi-platform biomarkers using a combination of data-driven reference genes with non-parametric Fisher's exact test [\[see publication\]](#).
- Worked with collaborators from Mayo Clinic on prediction of clinical outcomes in oligometastatic castration-resistant prostate cancer (omCRPC) treated with SABR using tumor-derived extracellular vesicles [\[see publication\]](#).
- Worked with collaborators from Nebraska Children's on identification of potential biomarkers of infection in the CSF using a rat model of S. epidermidis catheter infection and proteomics [\[see publication\]](#).
- Developed novel graph-based databases for biological data integration, enabling efficient mining of complex cancer informatics data and omics data from extracellular vesicles [\[see publication\]](#).

Extracellular Vesicles in Prostate Cancer

- In collaboration with a renowned team of radiation oncologists and immunologists at Mayo Clinic, designed and implemented a calibrated, standardized flow cytometry platform for robust and reproducible measurement of extracellular vesicles (EVs), enabling translational prostate cancer (PCa) biomarker development [\[see publication\]](#).
- Developed transcriptomics and proteomics analyses to characterize EV cargo, supporting identification of clinically relevant biomarkers and therapeutic targets.
- Contributed to cross-functional research efforts bridging assay development, biomarker discovery, and clinical translation in PCa [\[see publication\]](#).

Microbiome in IBD

- In collaborations with colorectal cancer research group at the University of Nebraska Medical Center, provided complete bioinformatics analysis, starting from obtaining sequence data from Illumina's *BaseSpace* platform to identifying key RNA/protein molecules in colorectal cancer in mouse models. Specifically, conducted total RNA and 16S rRNA sequencing analysis to characterize tissue transcriptome and gut microbiome alterations in *Claudin* transgenic and knockout models and to study the role of dysbiosis in colorectal cancer [\[see publication\]](#).
- Developed a split-graph computational model to represent microbiome composition and interspecies interactions [\[see publication\]](#).
- Built a bioinformatics pipeline combining genomic data and pathway analysis to enable functional interpretation of microbiome dynamics in IBD [\[see publication\]](#).

BOAST-ABM: Automated Model Generation

- Created a novel platform using Abstract Syntax Trees (ASTs) to translate YAML specifications into executable Python code for agent-based modeling.
- Applied to public health setting for studying preferential attraction of Mosquitos on Dengue virus infected hosts and to cancer modeling for studying adaptive therapy [\[see publication\]](#).

Immuno-informatics

- In collaborations with Selin lab at UMass Chan Medical School, applied immuno-informatics and analysis of TCR repertoire sequencing to profile adaptive immune responses across conditions.
- Performed clonotype analysis to assess clonal expansion, repertoire diversity, and shared/public TCRs, generating insights into immune response dynamics relevant to infectious disease and immunotherapy [[see publication](#)].

Drug Informatics

- Performed large-scale screening of natural product databases to identify structurally similar fragments with potential pharmacological relevance [[see publication](#)].
- Identified enriched chemical motifs in antibacterial, antineoplastic, and cardiovascular drug classes, supporting discovery of bioactive compounds from natural sources [[see publication](#)].
- Explored multiple computational drug repurposing strategies to prioritize candidates for treatment of Ulcerative Colitis [[see publication](#)].

Technical Skills

- **Languages:** Python, R, Java, C/C++, Perl, Matlab.
- **Cloud & Infrastructure:** AWS, GCP, Azure, Docker, HPC (Slurm), Linux.
- **Bioinformatics Tools:** Nextflow, Snakemake, GATK, STAR, Seurat, QIIME 2, ChimeraX and many more.
- **Web & Database:** Shiny, Graph Databases (Neo4j), Java Web Apps, JS, SQL.

Education

- **Ph.D. in Biomedical Informatics** | University of Nebraska at Omaha | 2025
- **M.S. in Computer Science** | University of Nebraska at Omaha | 2009
- **B.E. in Computer Engineering** | Tribhuvan University, Nepal | 2005

Professional Memberships & Awards

- **Member:** International Society for Extracellular Vesicles.
- **Reviewer:** *BMC Bioinformatics*, *Journal of Computational Biology*, ISMB Conferences.
- **Award:** Key Personnel on NIH and Mayo Clinic research grants.

Certifications & Professional Development

- *Coursera certificates on a) Genomic and Precision Medicine, b) Virology I: How Viruses Work, c) Network Analysis in Systems Biology, d) Social Network Analysis, e) Startup Engineering, and f) Machine Learning*